$P_{total} \times V_{total} = 1 \text{ atm} \times 22.4\ l$

$\therefore P_{total} = \dfrac{1 \text{ atm} \times 22.4\ \cancel{l}}{5\ \cancel{l}}$

$P_{total} = 4.48 \text{ atm}$

$\therefore P_{O_2} = 0.21 \times 4.48 \text{ atm}$

$= 0.94 \text{ atm}$

$P_{He} = x_{He} \times P_{total}$

$x_{He} = \dfrac{n_{He}}{n_{O_2} + n_{He}}$

$= \dfrac{2.05}{0.54 + 2.05}$

$x_{He} = \dfrac{2.05}{2.59} = 0.79$

$\therefore P_{He} = 0.79 \times 4.48 \text{ atm}$

$P_{He} = 3.54 \text{ atm}$

6 (b) $2KClO_{3\,(s)} \longrightarrow 2KCl_{(s)} + 3O_{2\,(g)}$

$P_{total} = 772 \text{ mm Hg}$

$P_{H_2O} = 26.7 \text{ mm Hg}$

$P_{total} = P_{O_2} + P_{H_2O}$

$\therefore P_{O_2} = P_{total} - P_{H_2O}$

$P_1 = 26.7 \text{ mm Hg}$ $\quad T_2 = 295 \text{ K}$

$T_1 = 300 \text{ K}$ $\quad P_2 = ?$

$\dfrac{P_1}{T_1} = \dfrac{P_2}{T_2}$

$\Rightarrow P_2 = \left(\dfrac{P_1}{T_1}\right) T_2 = \dfrac{26.7 \text{ mm Hg}}{300\ \cancel{K}} \times 295\ \cancel{K}$

$P_2 = 26.26 \text{ mm Hg}$

$\therefore P_{O_2} = 772 - 26.26$

$= 745.74 \text{ mm Hg}$

7) $t_1 = 1.5$ minutes $\quad (gas)_{\text{hydro carbon}}$

$t_2 = 4.73$ minutes $\quad (gas)_{\text{Bromine}}$

$\dfrac{\gamma_{\text{Hydrocarbon}}}{\gamma_{\text{Bromine}}} = \dfrac{t_{\text{Bromine}}}{t_{\text{Hydrocarbon}}}$

($\because$ Volume is constant)

$= \dfrac{4.73 \text{ minutes}}{1.5 \text{ minutes}}$

$= 3.15$

$\dfrac{\gamma_{\text{Hydrocarbon}}}{\gamma_{\text{Bromine}}} = \sqrt{\dfrac{M_{\text{Bromine}}}{M_{\text{hydrocarbon}}}}$

$3.15 = \sqrt{\dfrac{159.8 \text{ gmol}^{-1}}{M_{\text{hydro carbon}}}}$

Squaring on both sides and rearranging,

$M_{\text{hydro carbon}} = \dfrac{159.8 \text{ g mol}^{-1}}{(3.15)^2}$

$M_{\text{hydro carbon}} = 16.1 \text{ g mol}^{-1}$

$n(12) + (2n+2)\ 1 = 16$ $\quad \left[\because \text{general formula for hydrocarbon } C_nH_{2n+2} \text{ (alkane)}\right]$

$12n + 2n + 2 = 16$

$14n = 16 - 2$

$14n = 14$

$n = 1$

$\therefore$ The hydro carbon is $C_1H_{2(1)+2} = CH_4$

8) Critical temperature of a gas is defined as the temperature above which it cannot be liquified even at high pressures.

$\therefore$ When cooling starts from 700 K, H_2O will liquify first, then followed by ammonia and finally carbondioxide will liquify.

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